

MailFlow

Dynamic Mail Transmission Solution Using Best Connectivity Across Modes

Deep Research Report — Crisis Numbers, Gaps & Winning Architecture

Problem Statement ID	SIH260461
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This report mirrors the depth of award-oriented SIH research: hard operational numbers, root-cause failure analysis, government pain points, competitive gaps, and a concrete two-person build plan so the team can ship a complete working prototype.

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1 Understanding India Post Transmission Reality

India Post operates the world's largest postal network. As of March 2025 it runs **1,64,999 post offices** (~90% rural), 23 Postal Circles, and a hub-and-spoke transmission backbone of National Sorting Hubs (NSHs), Intra-Circle Hubs (ICHs), Parcel Hubs, Transit Mail Offices (TMOs), Railway Mail Service (RMS), Mail Motor Service (MMS), and third-party modes (rail, air, road, water, occasional helicopter).

Network snapshot (2025)

Metric	Value
Total Post Offices	1,64,999 (rural ~1,49,385)
Sorting / processing facilities	1,100+ (NSH + ICH + Parcel + RMS units)
NSHs / ICHs (Speed Post processing)	~50 NSHs + ~211 ICHs (post-rationalisation under MPOP)
Parcel Hubs	~190 (L1 + L2) + 233+ Nodal Delivery Centres
Modes used for transmission	MMS road, hired road, RMS rail, air, surface, water

Mail Network Optimisation Project (MNOP, from 2010) and the newer Mail & Parcel Optimisation Project (MPOP) rationalised hubs and introduced common processing for Speed Post and Registered articles. Despite this, **routing decisions between hubs remain largely static and manual**. When a leg fails (cancelled flight, delayed train, blocked highway, security detention, embargo, weather), there is no system that instantly identifies affected consignments, recomputes best alternatives across all modes, and informs customers.

2 The Crisis — Hard Numbers That Expose the Problem

■■ **CRITICAL FINDING** — The transmission layer is the silent failure point. India Post processes enormous volume and handles lakhs of grievances, yet planners still lack a live multimodal decision engine.

1.65L+	14.3L	~90%	<1 day
Post Offices worldwide largest network	Complaints handled FY25 (DoP)	Rural share of network — only formal reach for many villages	Target for many Speed Post metro legs; still frequently missed in disruption

Grievance & volume pressure

Metric	Figure	Source implication
Complaints received FY24	~14.10 lakh	Scale of citizen friction
Complaints FY25	~14.33 lakh	No cases pending >180 days, but volume is structural
CPGRAMS (DoP, multi-year)	~1.83 lakh received (sample window)	DoP consistently among high-volume departments
Top pending circles (flagged)	MH, DL, UP, GJ, HR	Concentration of operational stress
Average resolution (internal)	~8 days (some longer)	Customer already experiences delay before complaint

Delay is the dominant complaint theme (late delivery, tracking gaps, surface transit black holes of 3–5 days, hub backlogs at major NSHs during peaks). Private couriers still outperform on speed for many urban corridors; India Post's unique strength is reach — which is exactly what fails when transmission is static.

Operational reality of transmission

- Hub rationalisation under MPOP is ongoing (NSHs downgraded to ICHs in phases) — network topology is still changing while planners use fixed plans.
- 80–85% of Speed Post originates from bulk centres in tier-1 cities; rural and remote legs remain capacity- and schedule-constrained.
- Set-to-set transfer delays inside mail offices have formal escalation rules (24h / 48h / 72h) — evidence that internal backlog is a known, managed risk, not an exception.
- Tracking gaps of 2–5 days on surface mail are treated as 'normal'; customers experience silence, not proactive ETA updates.
- Weather, rail punctuality (~73–80% in recent years for Mail/Express), air cancellations, and road blockages routinely break static routes with no automated blast-radius response.

3 Root Cause Analysis — Why the Current System Fails

Root Cause	What happens in practice	Impact
Static route plans	Mode/route chosen from fixed schedules, not live capacity, weather, or reliability history	Sub-optimal paths even on normal days
No disruption blast-radius engine	When a flight/train/road fails, identifying affected bags/consignments is manual and slow	Cascading delays across downstream hubs
Fragmented visibility	No single operational picture across road + rail + air + water + MMS	Planners cannot see 'what is at risk right now'
No learning loop	Rich daily transit history is not fed back into future edge weights	Same bad legs keep being chosen
Silent customer channel	ETA slip does not trigger proactive sender/addressee alerts with new ETA	Complaints after the fact; trust erosion
Capacity blindness	Purchased transport capacity utilisation is not visible per leg at planning time	Over/under booking; wasted spend
Adapter gap to live feeds	No public real-time hub/carrier capacity API; systems assume static schedules	Cannot close the loop with live carriers

These are not 'missing features' — they are the exact reasons the problem statement asks for: optimum mode/route using best connectivity + real-time data + space availability; interactive booking with transport agencies; monitoring + alternative routes + self-learning; and alerts on changes/delays.

4 What the Department of Posts Actually Needs

- A **planner-facing decision console** (not another public tracking website) that answers in real time: best route now, what is at risk, what broke, what to do.
- Time-dependent multimodal routing that respects capacity, mail class policy (Speed Post vs bulk), and live disruptions.
- One-click blast-radius identification + bulk re-route proposals with Δ ETA when any leg/hub fails.
- Automatic customer alerts (email/SMS-ready) and a public tracking link that reflects revised ETAs.
- Measurable reliability scores that improve future routing (learning loop).
- Auditability, embargo handling, bulk induction — because this is a government operations system.
- Architecture that can start on realistic simulation and later plug into real departmental/carrier feeds via adapters (no public live feed exists today).

5 What We Should Build to Win — MailFlow

MailFlow is a decision-support console for mail operations planners. It models the postal network as a **live, time-dependent multimodal graph** and runs the operational loop: induct → rank routes → reserve capacity → detect disruption → re-solve → alert → update reliability.

Core components mapped to PS requirements

PS Requirement	MailFlow Component
Predict optimum mode/route using best connectivity, real-time data, space availability	Time-expanded multimodal routing (Dijkstra) + capacity-aware edges + live disruption feed
Mode of transmission interactive with transport agency	Capacity & Bookings: request → confirm → manifest handshake
Interface for monitoring, alternative routes, self-learning	Control Tower, Live Map, Disruption Center, EWMA reliability feedback
Alerts to sender and addressee on changes/delays	Automated email (SMS-ready) + public tracking page per consignment

6 Unique Differentiators — Why We Win

- **Built for the real user** — operations planner console, not a consumer tracking site.
- **Classical, explainable algorithm** — time-dependent shortest path the team can defend line-by-line; every route has a plain-English rationale.
- **Honest self-learning** — EWMA reliability scores that visibly evolve from recorded performance (no black-box ML required for the prototype).
- **Disruption-first design** — What-If Simulator + blast-radius recompute treat failure as the normal case (exactly the Department's pain).
- **Deployment-ready thinking** — adapter architecture, audit trail, embargo rules, role-based access, bulk CSV induction.
- **Honest data strategy** — no fabricated live India Post feed; realistic simulated network (25 hubs, ~120 legs) + live weather/email; real feed = one new adapter.

7 Competitive / Existing Systems & Gaps

Existing capability	What it does	Gap vs MailFlow
India Post tracking (web/app)	Scan events, status history	Visibility only — no routing decision, no re-plan, no blast radius
MNOP / MPOP hub rationalisation	Network redesign, common Speed Post + Registered hubs	Structural optimisation, not dynamic per-consignment routing
Manual planning + fixed schedules	Planners use experience + static plans	Slow under disruption; no capacity-aware multi-objective ranking
Private logistics optimisers	Advanced for closed fleets	Not designed for India Post multimodal + public-service constraints + government audit
Generic OR-Tools / graph demos	Academic shortest path	No ops console, no disruption workflow, no learning loop, no alerts, no adapter story

Gap summary: **no existing public or departmental system closes the loop from live multimodal conditions → ranked routes → capacity booking → disruption re-plan → customer alert → reliability learning** inside one planner console.

8 Recommended Tech Stack

Layer	Technology
Routing engine	Pure TypeScript — time-expanded graph + Dijkstra; capacity edges; per-class weights; unit-tested
Frontend	React + TypeScript, Tailwind, Framer Motion, Recharts, Leaflet/OSM
Backend	Node.js + Express + TypeScript; REST + WebSocket
Data & auth	Supabase (PostgreSQL, RLS, role-based auth)
Live integrations	OpenWeatherMap (auto disruptions), Resend (email); SMS-ready adapter
Self-learning	EWMA of per-leg on-time performance → edge weights
Demo realism	Simulation clock + scripted disruption day (full ops day in ~5 min)

9 Key Presentation Data Points (use these on slides)

- 1,64,999 post offices — largest network on earth; ~90% rural — social infrastructure, not just logistics.
- 14+ lakh complaints/year — delay and tracking silence are structural, not anecdotal.
- Static plans + no blast-radius engine = cascading delays when any leg fails.
- MailFlow answers four questions in one console: best route now / at risk / what broke / what to do.
- Explainable Dijkstra on time-expanded multimodal graph — defensible in front of judges and the Department.
- Learning loop (EWMA reliability) + proactive alerts + public tracking = trust repair for citizens.
- Adapter architecture = honest about missing live feeds; ready for real departmental integration later.

10 Two-Person Complete Ownership Split + GitHub Plan

Work model: freeze the shared contract once (1–2 hours), then each person finishes their full side independently. Integrate only when both Definitions of Done are green. Push to GitHub on feature branches; merge via PRs.

PERSON 1 — Full Backend + Routing Brain + Intelligence

- packages/shared-types + docs/api.md (with Person 2, then freeze)
- packages/routing-engine: time-expanded graph + Dijkstra, capacity edges, class weights, unit tests
- apps/api: Express+TS, consignments, routes, capacity/bookings, disruptions, reliability, embargo, bulk CSV, audit
- WebSocket events for positions + disruptions
- Supabase schema + seed (25 hubs, ~120 multimodal legs, delay distributions)
- EWMA reliability, deadline-risk scoring, What-If / blast-radius re-solve
- OpenWeatherMap + Resend adapters; simulation clock + scripted 5-min full-day disruption scenario
- README, env docs, seed scripts — API fully demoable via Postman/curl alone

PERSON 2 — Full Frontend Console + Map + UX + Demo Polish

- apps/web: React+TS+Tailwind+Framer+Router; dark blue + orange design system
- Control Tower KPIs + activity feed
- Route Planner (3 ranked options + English rationale + confirm)
- Live Network Map (Leaflet): hubs, legs by mode, moving consignments, disruption overlays
- Disruption Center + What-If Simulator UI
- Consignments register, Analytics (Recharts), public tracking page /track/:token
- Admin: embargo, bulk CSV, audit trail viewer
- Mock layer matching shared-types so the entire UI is demoable before real API exists
- Presentation mode, empty/loading/error states

GitHub rules

- Monorepo: packages/shared-types, packages/routing-engine, apps/api, apps/web, data/, docs/
- After contract freeze: Person 1 only touches api + routing-engine; Person 2 only touches web (+ mocks)
- Feature branches → PR into main; no direct push to main after Day 1
- Conventional commits; every PR must typecheck
- Integrate only when both sides pass their own Definition of Done

11 Implementation Roadmap (Independent → Integrate)

Phase	Who	Outcome
Contract freeze (1–2 h)	Both	shared-types + api.md locked
Person 1 complete backend	P1 alone	All API + engine + simulation green on Postman
Person 2 complete frontend	P2 alone	All screens green on mocks
Integration	Both	Real API + WS wired; full loop works
Deploy + 5-min demo	Both	Live URLs + recorded fallback video

With Antigravity / AI agents assisting both persons, expect ~2.5–4 focused days per person for their full side, then 1–2 days integration. Total calendar ~4–6 days when parallel.

12 Open Questions & References

Open questions to clarify with mentors / DoP contacts (if available)

- Preferred pilot corridor or circle for demo narrative?
- Any internal non-public schedule/capacity APIs the team could later adapter to?
- Mail-class weight policy defaults the Department would recognise?

Primary sources used

- Department of Posts network & MPOP/MNOP public OMs and parliamentary replies (NSH/ICH counts, hub rationalisation).
- Standing Committee / DoP grievance figures (FY24 ~14.10L, FY25 ~14.33L complaints).
- CPGRAMS department-wise receipt data; India Post tracking & delay pattern documentation.
- Rail punctuality series; India Post annual/network statistics (1,64,999 POs, rural share).
- Problem statement SIH260461 expected solution mapping (as captured in project report).

MailFlow · SIH260461 · Deep Research for an Award-Oriented Build · Integrate only when both sides are complete